MeatCODE - the simple architecture

26 October 2026 · Design for review, not yet implemented.

Two connected flows: turn papers into reliable evidence, then use that evidence and relevant user context to answer questions.

Build the knowledge

• **1. Scientific papers:** The source material, with each finding linked back to its paper.

• **2. GROBID:** Converts PDFs into structured text for extraction.

• **3. M1 / Qwen:** Runs locally on the PC and proposes entities, relationships and claims.

• **4. Validation / M2 + human:** Checks the extracted claims against their evidence. Humans approve initially; reviewed examples can support later model training and evaluation.

• **5. Knowledge / Neon:** Stores approved evidence and its source links. Active evidence is searched normally; valid but lower-priority evidence stays in reserve. Unresolved claims remain in review.

Answer and adapt

• **6. Oracle + API:** Receives the question, brings in relevant user context and coordinates the answer. The API is grouped with Oracle to simplify the map.

• **7. Hybrid retrieval:** Selects useful evidence through keywords, vector similarity and graph connections. Sends a limited evidence package to GPT. Reserve-search rules remain to be decided.

• **8. M3 / GPT:** Reasons over the selected evidence and user context, then returns a cited answer through the API. It identifies uncertainty when evidence is insufficient.

• **9. User memory / Neon:** Saves useful goals, preferences, constraints and corrections. Updates this memory after interactions and uses relevant parts in later questions. Users can review, change or delete their memory.

The repeating loop

Ask → retrieve → answer → feedback → update memory → ask again.

• Memory helps choose more relevant evidence and suggestions; it does not automatically retrain GPT or execute suggested actions.

• User statements remain separate from verified scientific evidence. Preferences must not hide contradictory findings.

• Logs, memory processing, artifact storage and ranking still exist behind the scenes. They are grouped within the nine objects rather than shown as separate steps.